Slide Set #3

Least Square Linear Fitting Problem (LS-LFP)

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Notes

Common data sets' format

	wage	enroll	kww	iq	married	libcrd14	exper	expersq
1	548	0	15	NA	1	0	16	256
2	481	0	35	93	1	1	9	81
3	721	0	42	103	1	1	16	256
4	250	0	25	88	1	1	10	100
5	729	0	34	108	1	0	16	256
6	500	0	38	85	1	1	8	64
7	565	0	41	119	1	1	9	81
8	608	0	46	108	1	1	9	81
9	425	0	32	96	4	0	10	100
10	515	0	34	97	1	1	11	121
11	225	0	29	84	1	0	13	169
12	400	0	34	89	1	1	8	64
13	417	0	22	93	1	1	7	49
14	217	0	27	74	1	0	13	169
15	894	0	43	116	1	1	9	81
16	300	0	36	NA	6	1	4	16

- Rows corresponds to measurements (sample units)
- Columns correspond to variables/features

Notes

Formal description of the data set

$\boldsymbol{y} \in \mathbb{R}^n$: column vector of the dependent/response/outcome variable (e.g wage)

$$\boldsymbol{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$\boldsymbol{x}_i \in \mathbb{R}^K$: column vector containing the K regressors (eg. years of school, dad's schooling, gender, etc) measured on the ith sample unit

$$\boldsymbol{x}_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{iK} \end{pmatrix}$$

Notes

So called design matrix, is the $(n \times K)$ matrix whose rows contain sample units and columns contain regressors

$$\boldsymbol{X} = \begin{pmatrix} x_{11} & x_{12} & \dots x_{1K} \\ x_{21} & x_{22} & \dots x_{2K} \\ \vdots & \vdots & \ddots \vdots \\ x_{n1} & x_{n2} & \dots x_{nK} \end{pmatrix} = \begin{pmatrix} \boldsymbol{x}_1{}' \\ \boldsymbol{x}_2{}' \\ \vdots \\ \boldsymbol{x}_n{}' \end{pmatrix}$$

Indexes and constants:

- n: number of sample units
- K: number of covariates/regressors/features measured on each of the n sample units
- $i = 1, 2, \dots, n$: indexes sample units
- $k = 1, 2, \dots, K$: indexes regressors

Notes

The linear fitting/approximation problem

In principle this is not necessarily a statistical problem. For the time being we consider $\boldsymbol{y}$ and $\boldsymbol{X}$ as a bunch of fixed data measurements.

Take the i-th measurement, and suppose we want to approximate the measurement y_i as a linear function of the measurements $\boldsymbol{x}_i$, that is

$$y_i \approx b_1 x_{i1} + b_2 x_{i2} + \cdots b_K x_{iK}$$

based on a set of K coefficients $\boldsymbol{b} = (b_1, b_2, \ldots, b_K)'$

What do we mean by ($\approx$)? Let's say we are willing to accept an approximation residual (a numerical mistake) e_i such that

$$y_i = b_1 x_{i1} + b_2 x_{i2} + \cdots b_K x_{iK} + e_i$$

Notes

But here we want more than this. We would like to have a sequence of n different residuals $\{e_1, e_2, \ldots, e_n\}$ such that the same coefficients work for all n measurements, that is the fallowing equations hold

$$\begin{aligned} y_1 &= b_1 x_{11} + b_2 x_{12} + \cdots b_K x_{1K} + e_1 \\ y_2 &= b_1 x_{21} + b_2 x_{22} + \cdots b_K x_{2K} + e_2 \\ &\vdots \\ y_n &= b_1 x_{n1} + b_2 x_{n2} + \cdots b_K x_{nK} + e_n \end{aligned}$$

The previous system can be written in matrix form as

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{b} + \boldsymbol{e}$$

where $\boldsymbol{e} = (e_1, e_2, \ldots, e_n)'$ is the residual vector

Notes

Good approximation

It is good if it guesses right, that is, it produces a small residual

The liner fitting problem (1)

is to find a $\boldsymbol{b}$ such that we get a *small residual*

Which residual?

at the the end of the day we will have n residuals $\{e_1, e_2, \ldots, e_n\}$

The liner fitting problem (2)

is to find a $\boldsymbol{b}$ such that we get an *overall small residual*

Yes, but how to measure the residual *overall*?

depending on how we measure the size of $\boldsymbol{e}$ we have different fitting methods.

Notes

Least squares fitting

Size of the residual

one simple way to measure the size of the residual is to use the (squared) Euclidean norm of $\boldsymbol{e}$

$$\|\boldsymbol{e}\|_2^2 = \boldsymbol{e}'\boldsymbol{e}$$

Cost/objective function

given that

$$\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}$$

define

$$S(\boldsymbol{b}) := \|\boldsymbol{e}\|_2^2 = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}).$$

The approximation is good if we make $S(\boldsymbol{b})$ as small as possible

Note that we can rewrite the least squares criterion

$$S(\boldsymbol{b}) = \boldsymbol{e}'\boldsymbol{e} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \boldsymbol{x}_i'\boldsymbol{b})^2$$

Notes

LS-LFP solution

is the vector of coefficient $\boldsymbol{b}$ that solve the following optimization program

$$\boldsymbol{b} := \underset{\boldsymbol{t}\in\mathbb{R}^K}{\arg\min}\ S(\boldsymbol{t})$$

Convexity

The LS-LFP objective function is convex (why?), therefore we can expect that $\boldsymbol{b}$ exists and is unique (why?)

Quality of approximation

the fact that $\boldsymbol{b}$ exists, doesn't tell us how small $S(\boldsymbol{b})$ will be. This depends on the input data $(\boldsymbol{y}, \boldsymbol{X})$

Notes

Proposition (existence)

Assume that the $(n \times K)$ matrix X has rank K. There exists a unique solution to the LS-LFP fitting problem that is

$$\boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$$

To see this, first we introduce two simple matrix derivative rules:

1. Let $\boldsymbol{a}, \boldsymbol{b} \in R^p$ then
$$\frac{\partial \boldsymbol{a}'\boldsymbol{b}}{\partial \boldsymbol{b}} = \frac{\partial \boldsymbol{b}'\boldsymbol{a}}{\partial \boldsymbol{b}} = \boldsymbol{a}$$
2. Let $\boldsymbol{b} \in \mathbb{R}^p$, and let $\boldsymbol{A} \in \mathbb{R}^{p\times p}$ be symmetric, then
$$\frac{\partial \boldsymbol{a}' A\boldsymbol{b}}{\partial \boldsymbol{b}} = 2\boldsymbol{A}\boldsymbol{b} = 2\boldsymbol{b}'\boldsymbol{A}$$

Notes

Proof. Rewrite the LS-LFP objective function

$$\begin{aligned} S(\boldsymbol{t}) =& (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{t})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{t}) \\ =& \boldsymbol{y}'\boldsymbol{y} - \boldsymbol{t}'\boldsymbol{X}'\boldsymbol{y} - \boldsymbol{y}'\boldsymbol{X}\boldsymbol{t} + \boldsymbol{t}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{t} \end{aligned}$$

Note that the transpose of a scalar is the scalar itself, then

$$\boldsymbol{y}'\boldsymbol{X}\boldsymbol{t} = (\boldsymbol{y}'\boldsymbol{X}\boldsymbol{t})' = \boldsymbol{t}'\boldsymbol{X}'\boldsymbol{y}$$

so that we write

$$S(\boldsymbol{t}) = \boldsymbol{y}'\boldsymbol{y} - 2\boldsymbol{t}'(\boldsymbol{X}'\boldsymbol{y}) + \boldsymbol{t}'(\boldsymbol{X}'\boldsymbol{X})\boldsymbol{t} \tag{3.1}$$

Since $S(\cdot)$ is convex, there exists a minimum $\boldsymbol{b}$ which will satisfy the first order conditions

$$\left.\frac{\partial S(\boldsymbol{t})}{\partial \boldsymbol{t}}\right|_{\boldsymbol{t}=\boldsymbol{b}} = \boldsymbol{0}$$

Notes

By applying the previous derivative rules (1) and (2) to the 2nd and 3rd term of (3.1)

$$\frac{\partial S(\boldsymbol{b})}{\partial \boldsymbol{b}} = -2(\boldsymbol{X}'\boldsymbol{y}) + 2(\boldsymbol{X}'\boldsymbol{X})\boldsymbol{b} = \boldsymbol{0}$$

Which lead to the so called "normal equations"

$$(\boldsymbol{X}'\boldsymbol{X})\boldsymbol{b} = \boldsymbol{X}'\boldsymbol{y}$$

$\boldsymbol{X}'\boldsymbol{X}$ is a square symmetric matrix. Since $\text{rank}(\boldsymbol{X}) = K$, then $\boldsymbol{X}'\boldsymbol{X}$ is non singular (see homeworks). Therefore $(\boldsymbol{X}'\boldsymbol{X})^{-1}$ exists, and using the normal equation

$$(\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{X})\boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} \quad \Longrightarrow \quad \boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$$

which proves the desired result ■

Notes

Formulation in terms of sample averages

It can be shown (see homeworks) that

$$\boldsymbol{X}'\boldsymbol{X} = \sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i' \qquad \text{and} \qquad \boldsymbol{X}'\boldsymbol{y} = \sum_{i=1}^{n} \boldsymbol{x}_i y_i$$

Define

$$\boldsymbol{S}_{\boldsymbol{xx}} = \frac{1}{n}\boldsymbol{X}'\boldsymbol{X} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i' \qquad \text{and} \qquad \boldsymbol{s}_{\boldsymbol{xy}} = \frac{1}{n}\boldsymbol{X}'\boldsymbol{y} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i y_i$$

Therefore $\boldsymbol{b} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$ can be written as

$$\begin{aligned}
\boldsymbol{b} &= \left(\frac{1}{n}\boldsymbol{X}'\boldsymbol{X}\right)^{-1} \frac{1}{n}\boldsymbol{X}'\boldsymbol{y} \\
&= \left(\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i'\right)^{-1} \left(\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i y_i\right) \\
&= \boldsymbol{S}_{\boldsymbol{xx}}^{-1}\, \boldsymbol{s}_{\boldsymbol{xy}}
\end{aligned}$$

Notes

Residuals

once $\boldsymbol{b}$ is fitted the residual is computed as

$$\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}$$

Fitted vector $\hat{\boldsymbol{y}}$

$$\boldsymbol{y} \;=\; \hat{\boldsymbol{y}} + \boldsymbol{e} \;=\; \boldsymbol{X}\boldsymbol{b} + \boldsymbol{e}$$

For all $i = 1, 2, \ldots, n$, the approximation of y is

$$\hat{y}_i = b_1 x_{i1} + b_2 x_{i2} + \ldots + b_K x_{iK}$$

Including an intercept/constant term

in some cases we want that b_1 works like an intercept for a straight line. This case is as if we fix $x_{i1} = 1$ for all $i = 1, 2, \ldots, n$ (the first column of $\boldsymbol{X}$ is filled with 1)

Notes

Algebraic/Geometric properties of the LS-LFP solution

Proposition (orthogonality of residuals)

The column space of $\boldsymbol{X}$ is orthogonal to the residual vector

Proof. Write the normal equations

$$\boldsymbol{X}'\boldsymbol{X}\boldsymbol{b} - \boldsymbol{X}'\boldsymbol{y} = \boldsymbol{0} \implies \boldsymbol{X}'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{b}) = \boldsymbol{0} \implies \boldsymbol{X}'\boldsymbol{e} = \boldsymbol{0}$$

Therefore for every column $\boldsymbol{X}_{\cdot k}$ (observed regressor) it holds true that the inner product $\boldsymbol{X}_{\cdot k}{}'\boldsymbol{e} = \boldsymbol{0}$. ■

Notes

Proposition (the residuals sum is zero)

If the linear approximation includes the constant term (intercept), then

$$\sum_{i=1}^{n} e_i = \sum_{i=1}^{n} (y_i - \boldsymbol{x}_i'\boldsymbol{b}) = 0$$

Proof. By assumption we have a liner approximation with a constant/intercept term.That is

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \ldots + e_i$$

Therefore $\boldsymbol{X}_{\cdot 1} = \boldsymbol{1}_n = (1, , 1, \ldots, 1)'$. Apply the previous property the 1st column of $\boldsymbol{X}$

$$\boldsymbol{X}_{\cdot 1}{}'\boldsymbol{e} = \boldsymbol{1}'\boldsymbol{e} = \sum_{i=1}^{n} e_i = 0$$

and this proves the property ■

Notes

Proposition (the fitted vector is a projection)

$\hat{\boldsymbol{y}}$ is the projection of $\boldsymbol{y}$ onto the space spanned by columns of $\boldsymbol{X}$

Proof.

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{b} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} = \boldsymbol{P}\boldsymbol{y}$$

It suffices to show that that $\boldsymbol{P} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ is symmetric and idempotent.

$$\begin{aligned}
\boldsymbol{P}' &= \left(\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\right)' \\
&= \boldsymbol{X}\left((\boldsymbol{X}'\boldsymbol{X})^{-1}\right)'\boldsymbol{X}' \\
&= \boldsymbol{X}\left((\boldsymbol{X}'\boldsymbol{X})'\right)^{-1}\boldsymbol{X}' \\
&= \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' = \boldsymbol{P}
\end{aligned}$$

Therefore $\boldsymbol{P}$ is symmetric.

Notes

$$\begin{aligned}
\boldsymbol{P}\boldsymbol{P} &= \left(\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\right)\left(\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\right) \\
&= \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{X})(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \\
&= \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' = \boldsymbol{P}
\end{aligned}$$

which shows that $\boldsymbol{P}$ is also idempotent, and this completes the proof ■

$\boldsymbol{P}$ it's called the influence matrix, because measures the impact of the observed ys on each predicted $\hat{y}_i$.

Elements of the diagonal of $\boldsymbol{P}$ are called leverages, because are the influence y_i on the the corresponding $\hat{y}_i$

Notes

Proposition (orthogonal decomposition of the OLS fit)

The OLS solution decomposes $\boldsymbol{y}$ in the sum of two orthogonal components

$$\boldsymbol{y} = \hat{\boldsymbol{y}} + \boldsymbol{e} = \boldsymbol{P}\boldsymbol{y} + \boldsymbol{M}\boldsymbol{y}$$

Remark: orthogonality implies that the individual contributions of each term of the decomposition of $\boldsymbol{y}$ are somewhat *well identified*.

Proof. First notice that

$$\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}} = \boldsymbol{y} - \boldsymbol{P}\boldsymbol{y} = (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{y} = \boldsymbol{M}\boldsymbol{y}$$

where $\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{P})$. Therefore

$$\boldsymbol{y} = \hat{\boldsymbol{y}} + \boldsymbol{e} = \boldsymbol{P}\boldsymbol{y} + \boldsymbol{M}\boldsymbol{y}$$

It remains to show that $\hat{\boldsymbol{y}} = \boldsymbol{P}\boldsymbol{y}$ and $\boldsymbol{e} = \boldsymbol{M}\boldsymbol{y}$ are orthogonal vectors.

Notes

First note that $\boldsymbol{M}\boldsymbol{P} = \boldsymbol{P}\boldsymbol{M} = \boldsymbol{0}$, in fact

$$(\boldsymbol{I} - \boldsymbol{P})\boldsymbol{P} = \boldsymbol{I}\boldsymbol{P} - \boldsymbol{P}\boldsymbol{P} = \boldsymbol{0}$$

Moreover

$$\langle \boldsymbol{P}\boldsymbol{y}, \boldsymbol{M}\boldsymbol{y} \rangle = (\boldsymbol{P}\boldsymbol{y})'(\boldsymbol{M}\boldsymbol{y}) = \boldsymbol{y}'\boldsymbol{P}'\boldsymbol{M}\boldsymbol{y} = \boldsymbol{y}'\boldsymbol{P}\boldsymbol{M}\boldsymbol{y} = \boldsymbol{y}'\boldsymbol{0}\boldsymbol{y} = \boldsymbol{0}$$

and this completes the proof ■

$\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{P}$ is called the residual maker matrix because it maps $\boldsymbol{y}$ into $\boldsymbol{e}$. It allows to write $\boldsymbol{e}$ in terms of the observables $\boldsymbol{y}$ and $\boldsymbol{X}$.

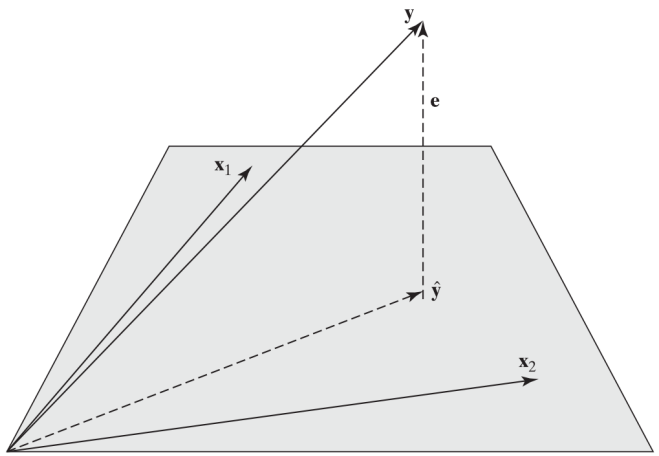
Properties:

- $\boldsymbol{M}$ is idempotent and symmetric (show it)
- $\boldsymbol{M}\boldsymbol{X} = \boldsymbol{0}$, in fact $\boldsymbol{M}\boldsymbol{X} = (\boldsymbol{I} - \boldsymbol{P})\boldsymbol{X} = \boldsymbol{X} - \boldsymbol{X} = \boldsymbol{0}$

Remark: it can be shown that this decomposition is also *unique* (a consequence of Hilbert projection theorem).

Notes

Least Squares Projection



Source: Greene, W. H. (2011) "*Econometric Analysis*" 7th Edition

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